

Back-Reference Mirror Recovery (BRMR): Field-Level Repair of Converter-Lost Owner Handles in DXF

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Abstract

Back-Reference Mirror Recovery (BRMR) rebuilds converter-lost owner handles in DXF files from the file's own surviving back-references. It stores each repair as a hashed sidecar in a storage manifest, never as an edit to the stored bytes. **Its answers on real files are corroborated, not scored:** the source DWGs have no independent reader here, so the 11 affected real files have no ground truth.

The error comes from ODA File Converter 27.1 (R2018 output). In 11 of 47 converted files, it writes 282 objects' owner handle (group 330) as 0, while each parent's 350/360 entry still lists the child. A census of all 31,688 mirrored objects found the back-reference agreeing with the owner every time. Each of the 282 orphans has exactly one parent pointing back.

The rule accepts an owner only when exactly one candidate passes a two-way check, and otherwise abstains. Under a sealed black-box protocol (frozen predictions, seeded injection, a single unseal):

- **Injected owner loss:** 555 of 555 fields restored; 36 of 36 files byte-identical to the hidden originals.
- **Adversarial mirrors:** an ambiguous back-reference is caught by uniqueness. A forged back-reference with no second witness defeats every file-only rule (115 of 115 wrong when answering). A second witness turns those errors into abstentions.
- **Random damage at 1% and 5%:** the file-only rule made 0 wrong answers. 7-8% of damaged owners were silent misses, landing on another real handle. An ingest-time Merkle commitment to the owner map (0.06% of corpus size) caught and repaired all 1,677.
- **Escalation:** escalating to a second witness on disagreement made 68% fewer errors than fixed priority (19 vs 59). Adding the parity witness gave 555 of 555.

On the real files, ODA's own audit, re-reading the source DWGs, assigned the same owner as BRMR to all 282 orphans. The same audit also rewrote 481 unflagged hatch geometry values, which supports storing the converter's output unchanged.

The principle of reading a reverse edge is prior art (fsck, the ezdxs auditor, CAD audit tools). The contribution is the abstain-unless-unique rule, the sealed evaluation protocol and the sidecar-only repair.

Naming.

- **History:** BRMR was developed under the working names GNP and FLMR. Repository files and the sidecar kind `flmr-owner-v1` keep those names, because their hashes are frozen and cited here.
- **Nickname:** BR0, "back-reference from zero". Every orphan it repairs starts as an owner of 0.
- **Not to be confused with** Lin et al.'s FLMR (multimodal retrieval, NeurIPS 2023) or Jiang et al.'s BRMR (temporal action localization, ICIG 2023).

1. Introduction

Converter errors can be undone from the error file alone, but only where the file mirrors the lost value, and only with a judge that a wrong answer would fail.

The DarkRock storage experiments needed a fixed DXF corpus. The 47 files came from real manufacturer DWGs, converted once with a pinned tool. That conversion is lossy: it rewrites group codes, respells numbers, shifts handles and leaves structural errors. The storage rule was set before any results: the restore target is the converter’s output, errors included, byte for byte. Repairing the files first would have measured restore of an improved file, not the stored one.

That rule left a question open, first raised by Grok: could a canonicalizer clean the bits back into place? Cleaning that replaces the stored bytes is a second conversion, so it is rejected. Rebuilding a separate view, without the original and with a measurable error rate, is a different claim, and it can be tested.

Reading a surviving reverse pointer to restore a lost forward pointer is not new. Filesystem checkers reattach files from directory listings, and DXF auditors repair or discard objects with invalid owners. This paper’s contributions are narrower:

1. **The error:** a characterization of the converter’s owner-handle error from its own output, and a census of every mirrored object in the corpus (§2).
2. **The rule:** an abstain-unless-unique repair rule stated over bytes, with a witness ladder that escalates only on disagreement (§3–4).
3. **The protocol:** a sealed black-box evaluation in which the method never sees the originals and each box is opened once, after a freeze (§5).
4. **Tests that can fail:** injected loss, adversarial mirrors, random damage and forged-mirror escalation, each with predictions frozen in advance and misses reported (§6).
5. **Sidecar repair:** repairs stored as hashed sidecars in a Reed–Solomon-protected manifest with a Merkle-committed owner map, so the stored bytes never change (§4.6, §8).

2. The converter error

Every one of the 282 reader repairs traces to one pattern: an object’s owner field is written as 0, while its parent still lists it as a child.

The corpus is 47 DXF files (R2018, AC1032) produced by ODA File Converter 27.1. An independent reader (ezdxf 1.4.4) audited them. It found 36 files with zero errors and zero fixes. The other 11 had zero errors but 282 in-memory fixes.

File family	Files	Object type	Fixes per file	Reader action	Surviving back-reference
081316 series-8150 sliding doors	4	TABLECONTENT	18	deleted	DATALINK → code 360
081476 sliding jamb options	3	XRECORD	2	owner repaired	DICTIONARY → code 350, plus the object’s own reactor
085313.20 vinyl patio doors	4	ACAD_PROXY_OBJECT	51	deleted	parent proxy → code 360

The raw bytes of one affected object show the pattern (CRLF line endings):

```

0
ACAD_PROXY_OBJECT
5
816B
330
0
100
AcDbProxyObject

```

An in-house scanner, written independently of the reader, finds exactly 18, 2 and 51 such orphans in these files, and none in the other 36. The error is therefore a lost forward pointer (child → parent). The reverse pointer (parent → child) is intact. That asymmetry is what BRMR exploits.

2.1 Mirror census

To check whether the back-reference can be trusted as a mirror, every non-root object in OBJECTS was classified in all 47 files, with no sampling and no injection:

Set	Mirror = {owner}	Ambiguous mirror	Mirror excludes owner	No mirror	Orphans
36 clean files	28,002	0	0	18	0
11 flagged files	3,686	0	0	20	282, each S = 1

Across 31,688 objects with a mirror, the mirror never named a different owner. The 38 objects with no mirror are all TABLEGEOMETRY, and the rule would abstain on them. This is a census of one converter’s output, not a property of the DXF format.

3. Model and notation

A file is a byte string, read as lines, then as objects. Every quantity below is defined on those bytes.

Split the file X on the newline byte into lines $\ell_0 \dots \ell_{m-1}$:

$$X = \ell_0 \parallel \backslash n \parallel \ell_1 \parallel \dots \parallel \ell_{m-1}, \quad |X| = \sum_{j=0}^{m-1} |\ell_j| + (m-1)$$

Lines pair into (group code, value). An **object** o is a maximal run of pairs that starts at code 0. For each object:

- **h(o)** is its handle (code 5 or 105).
- **$\lambda(o)$** is the index of its owner line: the first code-330 value outside a 102 {...} group and before the first code 100.
- **owner(o)** is ℓ at $\lambda(o)$, with any trailing carriage return removed.
- **ref(o)** is the set of (code, handle) pairs it points to, codes 320–369, 390, 480, 481 and 1005.
- **rx(o)** is the set of handles in its own {ACAD_REACTORS} group.

The handle universe and the candidate space for one owner are:

$$H(X) = \{h(o) : o \in X\}, \quad \Omega(o) = H \cup \{\perp\}$$

Here $\perp$ means “no answer” (abstain). In this corpus, |H| ranges from 289 to 28,308 per file, about 10,100 per injected field on average. The function $cr(\ell)$ returns the carriage return byte if ℓ ends with one, and the empty string otherwise, so line endings are preserved exactly.

4. Method: Back-Reference Mirror Recovery

BRMR finds each hole, reads the surviving mirror, and accepts an answer only if it is unique under a check that a wrong answer would fail.

4.1 Detection

The detector reads only the error file E:

$$D(E) = \{o \in \text{OBJECTS}(E) : o \neq \text{root}, \text{owner}(o) \in \{0, \emptyset\} \vee \text{owner}(o) \notin H(E)\}$$

4.2 The mirror identity

A DXF stores the ownership relation in both directions:

$$\text{owner}(o) = p \iff (350 \mid 360, h(o)) \in \text{ref}(p)$$

The converter erased the left side. The right side defines the solution set:

$$S(o) = \{ p \in H : (350 \mid 360, h(o)) \in \text{ref}(p) \}, \quad |S(o)| = 1 \Rightarrow \text{unique recovery}$$

If $|S(o)| = 0$, the information is gone. If $|S(o)| > 1$, it is ambiguous. In both cases the correct output is $\perp$.

4.3 Rebuilders

Each rebuilder proposes at most one candidate, so the Ω search is cut down to three proposals:

$$R_1(o) = \text{owner}(o_j), \quad j = \arg \min_j |i - j| \text{ over same-type objects with a valid owner} \quad (\text{array neighbor})$$

$$R_2(o) = \min_{\text{file order}} \{ p : (360, h(o)) \in \text{ref}(p) \}, \text{ else the same over 350} \quad (\text{back-reference})$$

$$R_3(o) = \text{first } r \in \text{rx}(o) \text{ with } r \in H \quad (\text{reactor})$$

R2 reads the mirror. R3 reads a partial mirror that some object types lack. R1 reads no mirror: it bets that neighbors share an owner.

R1 is a diagnostic only and is never in the accepting set. The GNP v1 version of R2 broke ties by file order. A3 (§6.6) shows that tie-break is unsafe, so the accepting rule is **R2u**, which abstains unless the mirror is unique:

$$R_{2u}(o) = \begin{cases} p, & S(o) = \{p\} \\ \perp, & |S(o)| \neq 1 \end{cases}$$

4.4 Selection rules

The priority order is $\pi = (R2 > R3 > R1)$, and first_π is the highest-ranked answer that is not $\perp$.

$$\begin{aligned} B(o) &= \begin{cases} v, & \#\{i : R_i(o) = v\} \geq 2 \\ \text{first}_\pi(R_2, R_3, R_1), & \text{otherwise} \end{cases} \quad (\text{control}+3) \\ S_{ij}(o) &= \begin{cases} R_i(o), & R_i(o) = R_j(o) \neq \perp \\ \perp, & \text{otherwise} \end{cases} \quad (\text{control}+2, \text{strict}) \\ F_{ij}(o) &= \begin{cases} S_{ij}(o), & S_{ij}(o) \neq \perp \\ \text{first}_\pi|_{\{i,j\}}, & \text{otherwise} \end{cases} \quad (\text{control}+2, \text{fallback}) \end{aligned}$$

There are two controls: **C0** leaves E unchanged, and **C1** deletes every object in D(E), the action the independent reader takes.

Ladder (escalate on disagreement). Ask the fewest witnesses that can settle the answer, and escalate only when they disagree. A known error is never available without the box, but a disagreement is visible. R4 is a type gate: a candidate passes only if its (parent type $\rightarrow$ child type) pair occurs in the clean corpus.

$$L(o) = \begin{cases} R_{2u}(o), & R_{2u}(o) = R_3(o) \neq \perp \quad (k = 2) \\ R_{2u}(o), & R_3(o) = \perp, R_4(R_{2u}(o)) \quad (k = 1) \\ R_3(o), & R_{2u}(o) \neq R_3(o), R_4(R_3(o)) \quad (k = 3) \\ \perp, & \text{otherwise} \end{cases}$$

Parity witness. At ingest, the manifest commits to the owner map as stored, with a Merkle root. A claimed owner is accepted only with a valid inclusion proof:

$$\text{leaf}(c, p) = H(00 \| c \| c \| p \| p), \quad \text{node}(l, r) = H(01 \| l \| r), \quad \text{verify}(\text{root}, \text{leaf}(o, p), \pi) \in \{0, 1\}$$

The prefixes and the root are those of RFC 9162 §2.1.1 (Certificate Transparency v2.0, which obsoletes RFC 6962). The tree is built bottom-up with an odd node promoted, which gives the same root as RFC 9162’s recursive split for every $n \geq 1$; a test checks $n = 1$ to 300. Only the empty tree differs: $H(0x02)$ here, $H(“ ”)$ in the RFC. The proof size is $O(\log n)$. The commitment protects against damage after ingest; it cannot repair errors made before ingest, because it commits to whatever was stored.

4.5 Rebuild (sidcar)

For any arm A , the rebuild writes each accepted answer into its owner line and changes nothing else:

$$\hat{X}_A = E[\ell_{\lambda(o)} \leftarrow A(o) \parallel \text{cr}(\ell_{\lambda(o)})]_{o \in D(E), A(o) \neq \perp}$$

The storage invariant is unchanged: $\text{STORE} = E$, and $\text{RESTORE}(\text{STORE}) = E$ byte for byte. The rebuild $\hat{X}$ is a separate view and never replaces the stored string.

4.6 Sidcar storage

In the DarkRock implementation (`src/manifest.rs`), each file’s `FileManifest` holds three layers that never mix:

Layer	Commits to	Purpose
<code>stripes + stripe_root</code>	RedTail-X 4+2 stripe manifests over the stored bytes	byte-exact restore, converter errors included
<code>owner_root</code>	the owner map as stored (Merkle root)	detects forged or silently changed owners after ingest
<code>sidecars</code>	one row per field: case, k used, each witness’s answer, decision; plus the view hash	the repaired view, replayable and hash-checked

Attaching or rendering a sidcar never changes `file_hash`, `stripe_root` or `owner_root`. A row whose expected value does not match the stored line is refused, so a sidcar can’t be replayed onto the wrong file.

5. Sealed black-box protocol

The method never sees the originals. The box is opened exactly once, after every rebuild exists, and any change to the method after opening requires a fresh box.

5.1 The box and the injected error

`BLACKBOX` = $\{X_1 \dots X_{36}\}$, the converted files with zero audit errors and zero fixes. The injection copies the converter’s own pattern. For seed s , with $k = \{\text{XRECORD: } 2, \text{TABLECONTENT: } 18, \text{ACAD_PROXY_OBJECT: } 51\}$, it samples $\min(k, \text{available})$ objects of each type that have a valid owner, giving the set I_s . It then rewrites one line per chosen object:

$$E = \text{ERR}_S(X) : \ell_{\lambda(o)} \leftarrow \emptyset \parallel \text{cr}(\ell_{\lambda(o)}) \quad \forall o \in I_S, \quad |X| - |E| = \sum_{o \in I_S} (\text{owner}(o) - 1)$$

Per seed, 555 fields were injected, and the box shrank from 117,392,559 to 117,390,661 bytes: 1,898 bytes erased. No other byte changed.

5.2 Sequence

1. **Freeze:** write the protocol and the SHA-256 of every script before any run.
2. **Seal:** inject the errors and hold the true owners $T(o)$ apart from the rebuild path.
3. **Rebuild blind:** detection, rebuilders and selection receive E only.
4. **Open once:** compare every rebuild with X .
5. **Stop:** the box is spent. A new method or a new rule needs a new seed.

Five boxes were used, each opened once: seed 20261001 (control+3), 20261002 (control+2), 20261003 (escalation, box v3), 20261004 (adversarial mirrors, A3) and 20261005 (random damage, A4). The script for the second run refuses to start if the frozen rebuild code has changed.

5.3 Scoring

For an arm A over the $N = 555$ injected fields:

$$C = \sum_{o \in I_S} \mathbf{1}[A(o) = T(o)], \quad W = \sum_{o \in I_S} \mathbf{1}[A(o) \notin \{T(o), \perp\}], \quad Ab = N - C - W$$

$$\text{precision} = \frac{C}{C + W}, \quad \text{coverage} = \frac{C + W}{N}$$

Because the injection touched only the lines $\lambda(o)$, a whole-file hash match is equivalent to every field in that file being correct:

$$\text{SHA256}(\hat{X}_A) = \text{SHA256}(X) \iff \forall o \in I_S \cap X : A(o) = T(o)$$

6. Results

The back-reference rebuilders decided the outcome: every arm containing it as a top-priority witness restored all 555 fields and all 36 files. Detection found 555 of 555 injected orphans, with 0 false detections, on both seeds.

The purge control C1 is not drawn. It deletes the orphaned objects, so it produces no owner values and 0 of 36 exact files. Best of three used a true two-of-three majority on 449 fields and fell back to R2 on 106.

6.1 By object type (correct of injected)

Type	n	R1 alone	R2 alone	R3 alone	Best of three
XRECORD	62	11	62	62	62
ACAD_PROXY_OBJECT	475	309	475	327	475
TABLECONTENT	18	0	18	0	18

Figures are from seed 20261001. TABLECONTENT has only 18 fields, so it is the thinnest cell.

6.2 Reproduction

The frozen script, rerun unchanged on a second machine (macOS, Python 3.10), produced identical totals, per-type counts and exploratory lines. Script hashes matched the freeze.

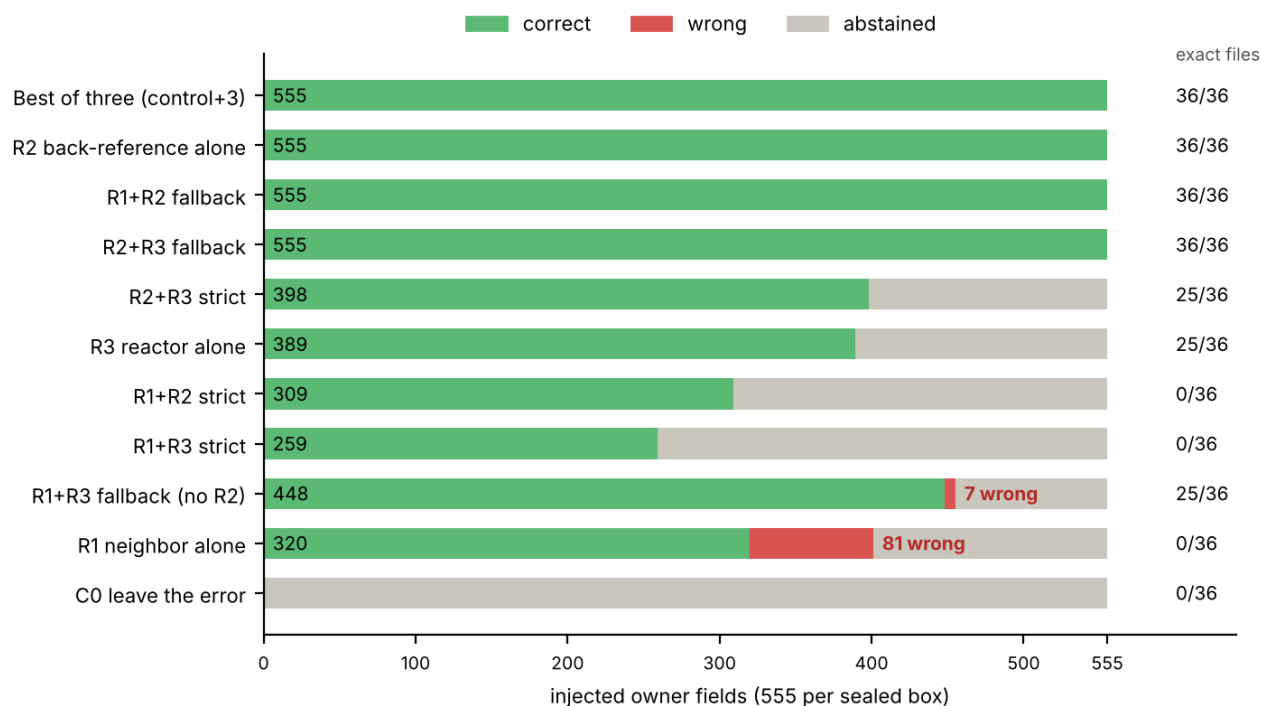


Figure 1: Correct, wrong and abstained per arm. Sources: gnp-blackbox-results.json (seed 20261001) and gnp-pairs-results.json (seed 20261002).

6.3 The 11 real affected files (exploratory, no truth available)

Family	Files	Orphans per file	Rebuilder agreement	Orphans left after best of three	Reader fixes before → after
081316 series-8150	4	18	R1 ≠ R2, R3 silent	0	18 → 0 each
081476 sliding jamb	3	2	R2 = R3 ≠ R1	0	2 → 0 each
085313.20 vinyl patio	4	51	R1 ≠ R2, R3 silent	0	51 → 0 each

The last column is the independent reader (ezdxf 1.4.4) auditing the best-of-three rebuilds: 282 fixes before, 0 fixes and 0 errors after. This shows the rebuilds are structurally valid. It cannot show that they match the source DWG's owners, because no independent DWG reader was available.

6.4 A1: the real files under R2u

With R1 out of the accepting set and R2u in place of the v1 tie-break, all 282 real orphans were still accepted, with the same owners as best of three. **But 276 of them rest on a single witness:** the 72 TABLECONTENT and 204 proxy objects have no reactor. Only the 6 XRECORDS had a second, agreeing witness. R1 disagreed with the accepted owner on all 282.

6.5 A2: the converter's own audit as an external witness

The 11 source DWGs were converted again with the same pinned ODA 27.1 binary, once with its audit enabled and once without, as a control.

Witness	Objects kept	Same owner as R2u	Different owner	Deleted
ODA 27.1, audit on (re-reads the DWG)	282	282	0	0
ezdxf 1.4.4 auditor (reads the DXF)	6	6	0	276

That moves the 276 single-witness answers to two sources that read different inputs. ODA’s auditor is a second opinion from the vendor that produced the error, not independent ground truth.

Two side results:

- **The audit rewrote far more than the owners.** It changed 481 hatch geometry values (angles, radii, one coordinate) that no reader had flagged.
- **ODA is not byte-reproducible.** The audit-off control differed from the published files in two header timestamps per file and in some XRECORD payload handles. No owner line differed.

6.6 A3: adversarial mirrors

Objects in the clean files were given decoy or forged back-references (seed 20261004, 475 fields across 20 object types).

Case	n	Old tie-break	R2u	R2u, reactor must agree	Strict pair	Merkle verify	Merkle lookup
Decoy added (S = 2)	180	155 wrong	abstain	abstain	abstain	180 correct	180 correct
Mirror forged, no reactor	115	115 wrong	115 wrong	115 wrong	abstain	abstain	115 correct
Mirror forged, reactor present	180	180 wrong	180 wrong	abstain	abstain	180 correct	180 correct

Uniqueness stops ambiguous mirrors. A forged mirror with nothing in the file to contradict it defeats every file-only rule that answers; this is the method’s measured limit. A second witness turns the error into an abstention. The committed owner map resolves all three cases.

6.7 A4: random damage

Every handle value in OBJECTS was replaced with random hex of the same length, independently at rate p (seed 20261005).

	$p = 0.01$	$p = 0.05$
Owner fields hit	280	1,397
File-only: wrong answers	0	0
File-only: abstained	6	61
File-only: silent misses (owner landed on another real handle)	19 (6.8%)	110 (7.9%)
Parity witness: caught and repaired	280	1,397
Parity witness: false flags	0	0

Random damage never produced the coordinated pattern that fools the rule. The file-only weakness under random damage is the silent miss, not the wrong answer. The prediction for silent misses (1–5%) was too low, because it used the median file’s handle density; dense files with short handles dominate.

6.8 Box v3: escalation against forged mirrors

555 erased fields, 10% of them forged (seed 20261003).

Arm	Correct	Wrong	Abstain
Priority (R2u, else R3)	496	59	0
Strict pair (R2u = R3)	354	0	201
Ladder	536	19	0
Ladder + parity	555	0	0

The ladder's 19 errors are exactly the forged fields with no reactor, 68% fewer than priority's. The type gate caught none, because every decoy had the true parent's type.

6.9 Sidecars in the storage manifest

The 282 A1 repairs were written as flmr-owner-v1 sidecars for the 11 real files:

- **Restore:** 25 RedTail-X stripes over 20,067,590 bytes, each restored with two of six shares missing.
- **Commitments:** an owner map of 3,988 edges committed as stored.
- **Views:** each rendered view differs from the stored file only on the accepted owner lines, and is byte-identical to the best-of-three rebuild that ezdx audits with 0 fixes. file_hash, stripe_root and owner_root were unchanged in all 11.

7. Analysis

What decided the results was the quality of the witnesses, not how many there were. Four short lemmas account for every row of the results chart.

7.1 Lemmas

Lemma 1 (strict pairs err only on a coincidence). A strict pair answers wrong only when both members give the same wrong handle:

$$W(S_{ij}) \leq \#\{o : R_i(o) = R_j(o) \notin \{T(o), \perp\}\}$$

Since $W(R2) = W(R3) = 0$, every strict pair scored $W = 0$.

Lemma 2 (strict coverage is the agreement rate). R2 was always correct, so R2 and Rj agree exactly when Rj is correct:

$$C(S_{2j}) = C(R_j) \Rightarrow C(S_{23}) = 398, C(S_{12}) = 309$$

All 157 abstentions of S_{23} are fields where R3 returned $\perp$: 18 TABLECONTENT and 139 ACAD_PROXY_OBJECT. R3 never disagreed with R2; it was only silent.

Lemma 3 (fallback collapses to its top member). If the top-priority member never abstains, a fallback pair equals that member:

$$Ab(R_2) = 0 \Rightarrow F_{12} = F_{23} = R_2$$

Best of three equals R2 unless $R1 = R3 \neq R2$, which cannot occur because $W(R3) = 0$.

Lemma 4 (a pair without the mirror inherits the guess). Without R2, the pair falls back to R1 wherever R3 is silent:

$$W(F_{13}) = \#\{o : R_3(o) = \perp, R_1(o) \notin \{T(o), \perp\}\} = 7$$

7.2 Why the neighbor rule fails

R1 bets that owner(o) equals its neighbor's owner. That holds for XRECORDs filed in one dictionary. It fails for proxy objects, which form parent → child chains where each child has its own owner. In all 7 errors, the true owner was the object's own parent (for example, 8114 is owned by 8113). R1 copied a neighboring chain's parent instead.

7.3 Weak and strong checks

A check's strength is the number of candidates it lets through:

$$\text{pass}(o) = \{p \in \Omega(o) : \text{check}(p)\}$$

- **Weak check ("the owner exists"):** $|\text{pass}| \approx 10,100$. It cannot separate a right answer from a wrong one. On the real files, two arms that disagree on 276 of 282 orphans both pass it.
- **Strong check ("p lists o as a child"):** $|\text{pass}| = 1$ on all 555 fields, on both seeds.

BRMR accepts only when $|\text{pass}| = 1$ under a check that a wrong candidate would fail.

7.4 Rule of three

A majority of n witnesses can correct f faulty ones only when:

$$n \geq 2f + 1$$

Three witnesses correct one fault. Two can only detect a disagreement, like a parity bit. In these runs the priority order placed the always-correct R2 first, so the fallback pairs matched best of three. Majority and priority would give different answers only where the top-priority witness is sometimes wrong. Box v3 (\$6.8) tests that case: with 10% of mirrors forged, escalating on disagreement made 19 errors, against 59 for fixed priority.

7.5 Erasure, not error

The detector locates each hole before recovery begins. That makes this erasure recovery, not error correction, matching the distinction in Reed-Solomon coding. An erasure at a known position needs one surviving copy. An error at an unknown position needs at least two redundant symbols.

7.6 How sure without opening the box

With zero errors in n independent trials, the 95% upper bound on the error rate is about 3/n:

$$\hat{p}_{95} \approx \frac{3}{n} = \frac{3}{555} \approx 0.54\%$$

That bound treats the 555 fields as independent. They are not: they sit in 36 files, come from one converter version and one injected pattern, and were all fields with a unique mirror. A systematic miss would fail a whole file at once, so the per-file bound is the more honest figure:

$$\hat{p}_{95}^{\text{file}} \approx \frac{3}{36} \approx 8.3\%$$

Both figures bound precision **given a unique owner-mirror**. Neither transfers to the 11 real files. The selection rule was frozen before the box opened, so the bound does not cover a choice made after seeing the result. The two seeds reuse the same files, so their trials are not pooled.

Recovery cost one pass per file to build a reverse index, then one lookup per field. A brute-force search with the strong check would have needed 5,605,588 candidate checks, and with the weak check it could not decide at all.

7.7 Random damage versus forgery

A single-witness answer is silently wrong only if three things happen together: the owner field is lost, the true parent's back-reference is lost, and some other reference becomes exactly $h(o)$. Random damage almost never does the third. It usually yields a handle that doesn't exist, so $S(o)$ becomes empty and the rule abstains. A4 measured 0 wrong answers in 1,548 detected hits.

Forgery is the opposite: it is built to make the third step happen, and A3 shows the rule cannot detect it from the file alone. The two damage models call for different defenses:

- **Random damage:** a single witness, escalating on disagreement, is safe for the answers it gives. The residual risk is silent misses, owners changed to another existing handle. Their rate rises with handle density: a random value of handle length hits an existing handle in 2.7% of cases in the median file, and in 80% in the densest.
- **Forgery, or any coordinated damage after ingest:** only a commitment made before the damage can tell. The owner map costs 82 KB compressed for 137.5 MB of DXF (0.06%), against about 50% for 4+2 Reed-Solomon parity. The Merkle roots alone are 1.5 KB.
- **Damage before ingest (the converter's own error):** no ingest-time commitment helps. Only external witnesses can add evidence: the converter's audit (\$6.5), or an independent DWG reader.

8. Relation to storage and prior practice

BRMR's closest relative is filesystem repair. RAID-1 is a loose analogy only: RAID-1 stores a second copy of the same data, whereas here the "second copy" is a different field that the format requires for its own reasons. BRMR complements deliberate redundancy rather than replacing it.

	Redundancy	Who added it	What it recovers
RAID-1 fsck / chkdsk	a whole second disk directory listings and inode back-links	the operator, on purpose the filesystem design	any block of a failed disk lost links between files and folders
DarkRock RedTail-X 4+2 Reed-Solomon	2 parity shares per stripe	the storage layer, on purpose	any bytes, up to 2 lost shares of 6
BRMR	the format's own parent ↔ child references	the DXF format, incidentally	only fields that still have a surviving copy

The closest prior practice is filesystem repair. A tool like fsck reattaches a file whose pointer to its folder is lost, because the folder's listing still names the file. R2 does the same with a parent object's child list.

CAD tools already perform the DXF-shaped version of this repair when a file is loaded:

- **ezdxf's auditor** reports `INVALID_OWNER_HANDLE`, then repairs the owner or discards the object; it discarded 276 of the 282 here.
- **ODA's own audit** reassigned all 282 (\$6.5).
- **Autodesk's RECOVER** command serves the same purpose.

What these tools don't do is abstain, record the evidence behind each change, or leave the stored file untouched. ODA's audit also changed 481 geometry values beyond the 282 owners. BRMR's difference is in those three properties, not in the idea of reading a back-reference.

Inside DarkRock, the two roles stay separate. The storage layer keeps the converter's bytes, errors included, and protects them with Reed-Solomon shares; byte-exact restore is judged against those stored bytes. BRMR produces an optional repaired view. It is judged against a sealed original when one exists, and against its own uniqueness and agreement checks when one does not.

The canonicalizer's control-plus-one rule (keep the control, or one verified alternative that restores exactly) needs no vote, because an exact answer is available. Rebuilding has no exact answer until the box is opened, which is why it needs witnesses and abstention.

9. Limitations and future work

The result is strong for one error class on one converter’s output, and silent about the source DWG.

Limitations

- **The real files are unscored.** The 282 real repairs agree with ODA’s own audit of the source DWGs, but that auditor comes from the vendor that made the error. No reader from an independent code-base was used.
- **Injected, not natural, errors.** The sealed boxes copy the converter’s pattern onto its own clean output. The real fault may differ in ways the injection doesn’t capture.
- **Forged mirrors are undetectable from the file alone** when no second witness exists (A3, box v3). The defense is an ingest-time commitment, which can’t help with errors made before ingest.
- **One error class.** Only owner-handle loss was tested. Respelled numbers, dropped records with no back-reference, and shifted handles were not; dropped records without a back-reference are unrecoverable in principle.
- **One converter version, reused files.** All boxes draw on the same 36 clean files, and the method’s design had already seen the converted corpus. The per-file bound (8.3%) reflects that better than the per-field bound.
- **Sparse cells.** TABLECONTENT contributes 18 injected fields in v1, and the type gate was tested only against same-type decoys.

Future work

1. An independent DWG reader (for example LibreDWG) as a third witness for the 11 real files.
2. Escalation beyond $k = 3$ with genuinely independent witnesses, using odd steps and a stop-on-margin rule rather than a fixed count.
3. The same protocol on other formats with built-in mirrors: PDF cross-reference tables and PNG chunk CRCs, using real files and fresh seeds.
4. Other error classes from the same converter, beginning with the hatch values its audit rewrites.

Glossary of symbols

Symbol	Meaning
X	the original file, sealed in the black box
$E = \text{ERR}_s(X)$	the error file: X with seed s’s owner lines set to 0; the stored object
$\hat{X}$	the rebuilt file, an estimate of X made from E alone; a sidecar view
s	random seed (20261001, 20261002)
ℓ_j, m	line j of a file; number of lines
$\text{cr}(\ell)$	the trailing carriage return of ℓ , if any
$o, h(o)$	a DXF object; its handle
$\text{owner}(o), \lambda(o)$	o’s owner handle; the line where it is written
$\text{ref}(p), \text{rx}(o)$	handles p points to; handles in o’s reactor group
H	every handle in the file, the universe of possible owners
$\Omega(o)$	candidate space $H \cup \{\perp\}$
$D(E)$	orphans detected in E
I_s	objects actually hit by the injection
$S(o)$	parents whose references list o
$R1, R2, R3$	neighbor, back-reference and reactor rebuilders
$\perp$	abstain, no answer
π	priority order $R2 > R3 > R1$
B, S_{ij}, F_{ij}	best of three; strict pair; fallback pair
$C0, C1$	controls: leave the error; delete orphans
A	an arm (any selection rule or control)
$T(o)$	true owner, read only when the box is opened
C, W, Ab	correct, wrong, abstained counts
$\text{pass}(o)$	candidates that satisfy a check
$n \geq 2f + 1$	witnesses needed to outvote f faulty ones

Added in this revision:

Symbol	Meaning
R2u	unique back-reference: the single p in S(o), else $\perp$
R4	type gate: passes a candidate whose (parent type $\rightarrow$ child type) pair occurs in the clean corpus
L	the ladder: escalate to more witnesses only on disagreement
owner_root	Merkle root of the owner map committed at ingest (the parity witness)
π (proof)	an inclusion proof, $O(\log n)$ hashes
ρ	random damage rate per handle value
sidecar	an flmr-owner-v1 record of repair rows plus the view hash; never changes the stored bytes

Reproducibility

All artifacts are in canon-test/GNP/ of the DarkRock repository:

- **Protocols and reports:**
 - GNP v1 and pairs: GNP-FREEZE.md, GNP-REPORT.md, GNP-PAIRS-FREEZE.md, GNP-PAIRS-REPORT.md;
 - A1, A3, A4: FLMR-A1-*, FLMR-A3-*, FLMR-A4-*;
 - A2: a2/FLMR-A2-*;
 - box v3: escalation/box-v3/FREEZE.md, REPORT.md;
 - sidecars: sidecar/SIDECAR-REPORT.md.
- **Code:**
 - run_gnp_blackbox.py, run_gnp_pairs.py, dxfobj.py, gnp_mirror_census.py;
 - run_flmr_real.py, run_flmr_a3.py, run_flmr_a4.py, escalation/box-v3/run_box_v3.py;
 - a2/run_oda_a2.sh, a2/compare_a2.py;
 - Rust: src/merkle.rs, src/manifest.rs, src/bin/flmr_sidecar.rs.
- **Results:** a JSON results file and a run log beside each report.
- **Math:** GNP-EQUATION.md, GNP-MATH-READER.md, GNP-GLOSSARY.md.

Frozen hashes (SHA-256):

8ef3f577fc8eceedfd48bf6a0537bf7004dc53a6c46972e6a7285911bb9df658	run_gnp_blackbox.py
cee72bd4956bc202eb50739f59e4f282909a77bf231f1a03c986642bbe3f633	dxfobj.py
07ee9da551e60b7617321f0f97396accb5e60b29a41dc134ffd06ed045b19e04	run_gnp_pairs.py
9e64689f5f568f113e33bf3e5595efa647d554179595f4b55c4c3d94a8c14e10	gnp_mirror_census.py
50125c43621fa3168c0b192896990710191de0ea36606c23fd3c71048e3d36c6	run_flmr_real.py
d0cda30bd15b1f853b266edb7236b31114c1790ef7af6ec796961e828d70ef8	run_flmr_a3.py
37b79174ae0af3d110c8d561d01989052c944ce7d6cfdc8bb0f5c45da3532240	run_flmr_a4.py
10197475d97f0207c6b39ee2956f8d5cc904faf5a0a5b88743a117a7bb84f729	run_box_v3.py
c55b7e84b955dfea865968f1a3a7d0e593aabb7e41eaf05770b97b65a3317355	run_oda_a2.sh
bc0b4a2d10fad02ba307d9d4bd696df39495379808c7d9e7f17751162f355a9f	compare_a2.py
be8937454323a7ca3afd34b31c439a540a3a67b6ebb3c654eae4fbb15265d2e9	src/merkle.rs
058b0158766e3f2b2f68b7a5b491b49db77c18ceef3fc0c65ff163e34dcac2cf	src/manifest.rs
3562ae456d198fcf0563d60181e3a16455acd898e129c1cc0828c7dba5ef5139	src/bin/flmr_sidecar.rs
4dc6691f4ce607f3722b2e6e35287e213901f45b4fec5a9aaf3302f5cf2a66b6	a2/ezdxf_owner_after_audit.py
ba1247d2278ee148306919d9a8fbada2bc0c803baab0f7977039f8b77116ebcb	sidecar/export_a1_rows.py
cd08298fecbbe65bf018a22764bac54d1dabb12d7ae084b478d6df151072b516	sidecar/verify_views.py

The input corpus is 47 DXFs converted by ODA File Converter 27.1 (R2018 output), manifest SHA-256 5ce26ceb2be36dfecdd43c929d0cd77a2f9331bd822fc7c426977ade66b139810. Independent reader: ezdxf 1.4.4. Each run takes about 24 seconds.

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All references were checked against their sources on 2026-10-02.